Evaluating Model Complexity and Generalization: Image Classification Strategies for Fashion-MNIST

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Summary

For this project, the Fashion-MNIST dataset looked to be the most interesting option provided to us, and it did not disappoint. To classify the images found within it, our team used a kNN classifier, logistic regression, a neural network classifier, and a random forest classifier. To achieve the maximum test accuracy, we implemented a neural network classifier with 128 hidden layers for a test accuracy of 88.14%.

Data Description

We used the Fashion-MNIST dataset for our experiments. This dataset was introduced by Zalando Research in 2017 as an alternative to the MNIST dataset, which consists of handwritten digits, to provide a more challenging classification problem. Fashion-MNIST was created using pictures of an assortment of fashion products from Zalando's online store. These pictures were then processed into standardized grayscale images to make them compatible with machine learning benchmarking.

The dataset contains 70,000 total images, with 60,000 of them used for the training set and the remaining 10,000 used for the test set. Fashion-MNIST was intended to keep the same structure as the MNIST digits dataset, maintaining 10 classes and an image size of 28 x 28, while also increasing the difficulty for machine learning models.

During our exploration of the dataset, we visualized sample images from different classes to better understand the types of clothing included. Figure 1 shows several images from the dataset. Although many of the clothing categories are visually distinct, some items, such as shirts, pullovers, and coats, appear similar, which makes them harder for classifiers to distinguish.

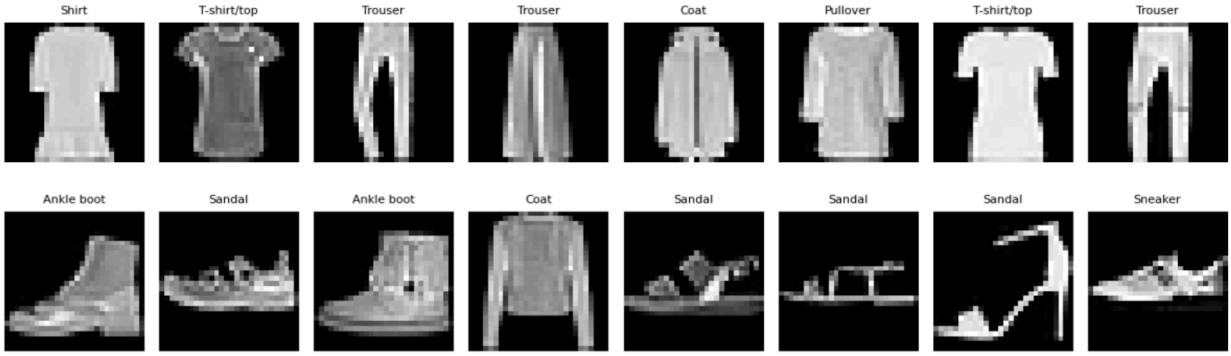


Fig. 1. Sample images from the Fashion-MNIST dataset showing clothing categories.

Classifiers

Classification 1: knn

K-nearest neighbours is roughly a memory-based classifier that predicts a data point's label by looking at the “k” most similar instances in the training set and comparing their Euclidean distances. In our implementation of KNN, the primary hyperparameter we investigated was k, the number of neighbors. We used k values {1, 3, 5, 7, 9, 11}, which helped us strike a balance between micro-level and macro-level trends.

Classification 2: Logistic Regression

Logistic Regression is, in essence, a linear model that converts weighted pixel inputs into class probabilities using the softmax function. When implementing this classifier, we use the “lbfgs” solver in scikit-learn, setting the maximum number of iterations (hyperparameter) to 1000 to ensure convergence of the parameters.

Classification 3: Neural Network (MLP)

This neural network uses a feedforward architecture with a 784-dimensional input layer, a hidden layer with ReLU activation, and a softmax output layer. In our implementation, we use scikit-learn to evaluate the model's capacity with varying hidden layer sizes (hyperparameters): {64, 128, 256}. Finally, we use stochastic optimization to minimize our cross-entropy loss, improving our overall performance.

Classification 4: Random Forest

Random Forest classification builds a collection of independent decision trees, each trained on a random bootstrap-sampled sample. In our implementation, we adjusted the forest size by varying the “n_estimators” (hyperparameter) over {10, 50, 100}, keeping “max_depth” at 15 to help prevent overfitting.

Experimental Setup

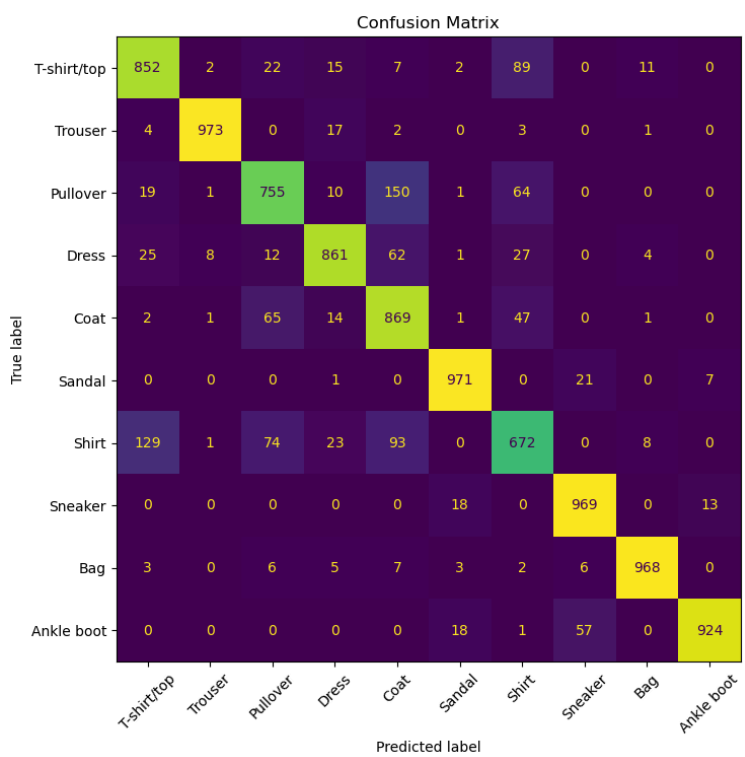
Each image from the Fashion-MNIST dataset was 28 x 28 pixels and belonged to one of ten classes. To prepare the data for classification, it was preprocessed and partitioned. For preprocessing the data, each 28x28-pixel image was flattened into a 784-dimensional vector with values between 0 and 255, representing the color intensity at each pixel. To create our data partitions for Fashion-MNIST, we split the data into sets of 10000 and 60000, following a 60/20/20 split. Where 60% of our data ≈ , 60,000 data inputs, were used for model training, another 20% as a validation set for tuning our hyperparameters, and the final 20% as a test set. Finally, we used accuracy as a metric to measure performance on the Fashion-MNIST dataset, which is balanced with around 1000 images per class.

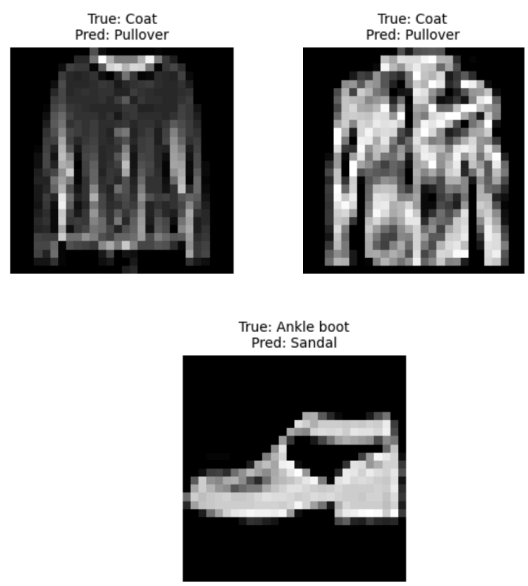
Experimental Results

Classifier	Best Hyperparameter	Validation Accuracy	Test Accuracy
kNN	k=5	82.3%	81.83%
Logistic Regression	*n/a*	85.19%	84.04%
Neural Networks (MLP)	hidden layers size = 128	88.4%	88.14%
Random Forest	n_estimator = 100	87.08%	86.39%

Insights

From this project, we observed that all of the classifiers performed well on the Fashion-MNIST dataset, with test accuracies ranging from 81.83% to 88.14%. After running all classifiers, the neural network (MLP) achieved the best performance based on accuracy alone. However, the overall difference in performance between the models was not extraordinarily large. While the neural network was clearly the most accurate, the other models still performed well, showing that even less complex classifiers can work effectively on the Fashion-MNIST dataset.





As it turns out, the MLP did not have a general misclassification problem. Instead, it heavily misclassified specific items of clothing for ones that were similar in nature. In the confusion matrix above, we can see that the MLP would misclassify pullovers for coats, or shirts for t-shirts, more than other items. Their shapes and textures appear to look alike, especially at the low 28 x 28 resolution, which makes it difficult even for humans to distinguish. Expectedly, the classifiers struggled with these images. We can also see that shirts were the most heavily misclassified items, with 328/1000 t-shirts being misclassified. Here are some examples of misclassifications above and to the right.

The Logistic Regression and Random Forest also had relatively good performance, considering the fact that they are not designed for image data. This suggests that the structure of the dataset allows models to achieve strong results even without deep architectural complexity. On the other hand, kNN had the lowest accuracy, likely due to the high dimensionality of the data (784 features), which makes Euclidean distance less effective at distinguishing between near and far neighbors. This understandably makes clustering less meaningful.

Considering the relative strengths and weaknesses of the models, the MLP had the best performance because it can model complex, non-linear relationships, allowing it to capture intricate patterns from the data, for example, the combination of pixels that show features like sleeves or collars. In contrast, Logistic regression is simpler to tune, but it is limited to linear relationships. kNN is also easy to implement, but as previously mentioned, it does not scale well in high-dimensional spaces.

If these models were applied to more realistic datasets, such as higher-resolution images or real camera data for object recognition, the differences in performance would likely become more apparent. More complex models, such as deep neural networks, which are well-suited to large-scale image recognition tasks, would likely outperform simpler models by a wide margin. Overall, this project showed that while model choice affects classification accuracy, even simple classifiers can perform well on structured image datasets such as Fashion-MNIST.

Contributions

Ryan - Implemented the knn, logistic regression, and MLP classifiers in code and gathered a report on the performance of each classifier.

Kyle - Completed the data description and many of the insights of the report

Rajvir - Implemented Random Forrest classifier. Wrote about each classifier in the report and contributed to the Experimental results table.

Greg - Wrote the summary and contributions section of the report, as well as a part of the insights